

Project-Low LevelDesign on Marketing Content Generator

Course Name: Gen AI

Institution Name:Medicaps University – Datagami Skill Based Course

Sr no	Student Name	Enrolment Number
1.	CHARU RATHOD	EN23CS3T1007
2.	RUDRAKSH SHARMA	EN22CS301832
3.	PRATHAM DESHMUKH	EN22CS301741
4.	PIYUSH PATIDAR	EN22CS301705
5.	ROHAN JAIN	EN22CS301817
6.	PRIYANSH MEWADA	EN22CS301763

Group Name: D7

Project Number:GAI-26

Industry Mentor Name: Aashruti Shah

University Mentor Name:Dr. Hemlata Patel

Academic Year: 2025-2026

Table of Contents

Chapter 1	Introduction
	1.1. Scope of the Document
	1.2. Intended Audience
	1.3. System Overview
Chapter 2	System Design
	2.1. Application Design
	2.2. Process Flow
	2.3. Information Flow
	2.4. Components Design
	2.5. Key Design Considerations
	2.6. API Catalogue
Chapter 3	References

1. Introduction

The Marketing Content Generator is a specialized Generative AI (GenAI) application designed to assist marketing professionals in producing high-quality, consistent, and professional content at scale. By leveraging sophisticated Prompt Engineering techniques, the system transforms basic user-provided topics or briefs into polished marketing outputs such as Ad Copy, product descriptions, social media posts, email campaigns, and taglines. The application ensures consistency in tone, formatting, and industry-standard terminology, significantly reducing the manual effort required for drafting complex marketing documentation.

1.1. Scope of the document

This document defines the Low-Level Design (LLD) of the Marketing Content Generator system.

It covers:

- Internal system architecture and GenAI integration design
- Component-level design of content generation modules
- Prompt Engineering pipeline and template management
- Data and process flow for content generation
- APIs and service structures

The goal is to provide a detailed technical blueprint for developers to implement and extend the system.

1.2. Intended Audience

The intended audience for the Marketing Content Generator includes marketing professionals, content strategists, copywriters, and product managers who need to produce large volumes of high-quality marketing content efficiently. It is also intended for developers and AI engineers responsible for building and maintaining the GenAI pipeline. The system is especially useful for organizations that want to standardize their marketing voice, reduce turnaround time for campaigns, and leverage AI to enhance creative output without replacing human creativity.

1.3. System overview

The Marketing Content Generator is a GenAI-powered system that:

- Accepts user inputs (topic, content type, tone, target audience)
- Applies Prompt Engineering techniques to construct optimized LLM prompts
- Sends engineered prompts to a Large Language Model (LLM) API
- Processes and formats the generated output for marketing use
- Supports multiple content types:
 - Ad Copy → persuasive advertising text
 - Social Media Posts → platform-specific content
 - Email Campaigns → subject lines and body copy

- Product Descriptions → SEO-optimized content
- Taglines → brand messaging
- Ensures consistent tone, style, and brand voice across all outputs
- Provides a user-friendly interface for marketing professionals

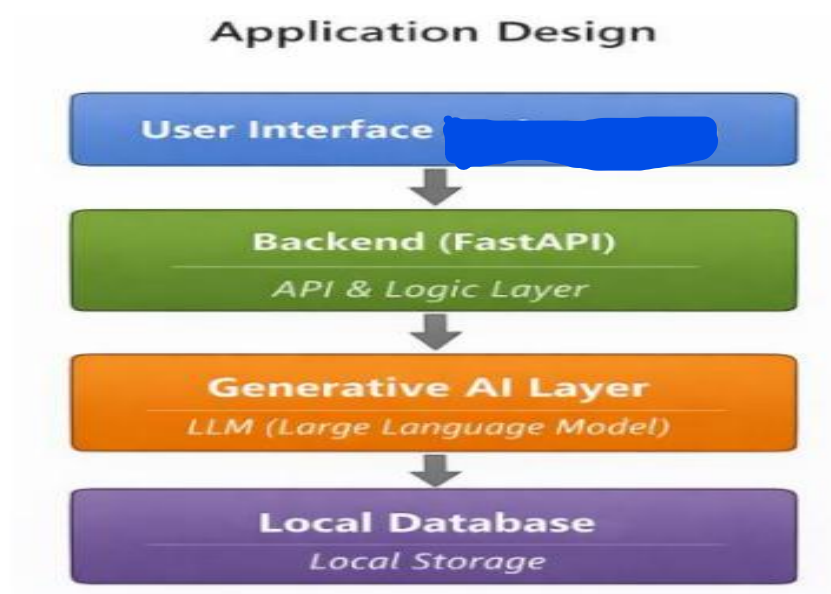
2. System Design

2.1. Application Design

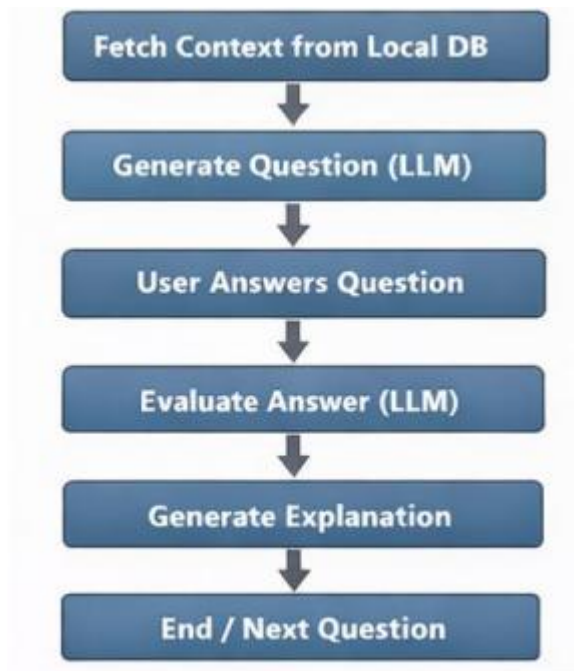
Architecture Type: Layered GenAI Application Architecture with Prompt Engineering Pipeline

Layers:

- Frontend (UI) → React (User input / content output display)
- Backend (FastAPI) → API handling + Prompt Engineering orchestration
- Prompt Engine Layer → Template management + dynamic prompt construction
- GenAI Layer → LLM API integration (OpenAI GPT / Google Gemini)
- Post-Processing Layer → Output formatting, tone validation, brand checks
- Data Layer → Database (PostgreSQL) for prompt templates and history



2.2. Process Flow



2.3. Information Flow

- Input → Topic, content type, tone, target audience, brand guidelines
- Intermediate → Engineered prompt, LLM raw response, formatting rules
- Output → Polished marketing content (Ad Copy, emails, posts, etc.)

2.4. Components Design

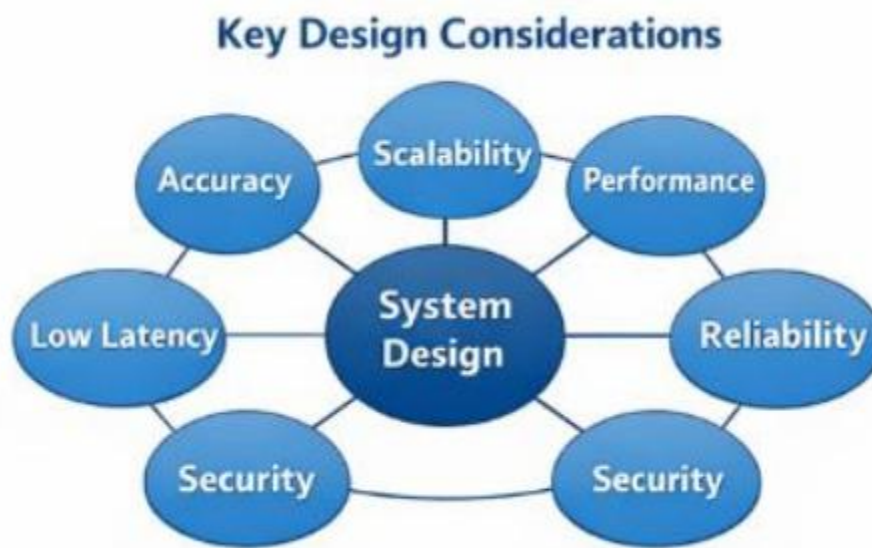
Components:

- Frontend UI
- API Controller (FastAPI)
- Prompt Engineering Engine (Core Brain)
- Prompt Template Manager
- Dynamic Prompt Builder

- LLM API Connector (OpenAI / Gemini)
- Post-Processor & Output Formatter
- Brand Voice Validator
- Database (PostgreSQL)

2.5. Key Design Considerations

- Scalability → Stateless API design allows horizontal scaling for high request volumes
- Modularity → Prompt templates and LLM connectors are independently swappable
- Fault Tolerance → Fallback to secondary LLM if primary API is unavailable
- Extensibility → New content types added by creating new prompt templates only
- Performance → Prompt caching reduces redundant LLM API calls
- Security → API key management via environment secrets; user data is never stored in LLM



2.6. API Catalogue

1. Generate Marketing Content

POST /api/generate

Request:

```
{
  "topic": "Summer Sale Shoes",
  "content_type": "ad_copy",
  "tone": "energetic",
  "audience": "young adults"
}
```

Response:

```
{
  "content": "Step into Summer! ■■■...",
  "content_type": "ad_copy",
  "tokens_used": 142
}
```

2. Get Content Types

GET /api/content-types

Response:

```
{
  "types": ["ad_copy", "email", "social_post", "tagline", "product_description"]
}
```

3. Get Generation History

GET /api/history?user_id=123

Response:

```
{
  "history": [
    { "id": 1, "topic": "Summer Sale", "type": "ad_copy", "created_at": "2026-04-01" },
    ...
  ]
}
```

Acti
Go to

3. References

- ☐ Generative AI Concepts and Large Language Models ☐ OpenAI API Documentation
- ☐ <https://platform.openai.com/docs>
- ☐ Google Gemini API Documentation
- ☐ <https://ai.google.dev>
- ☐ Prompt Engineering Guide
- ☐ <https://www.promptingguide.ai>
- ☐ FastAPI Documentation
- ☐ <https://fastapi.tiangolo.com>
- ☐ React Documentation
- ☐ <https://react.dev>
- ☐ PostgreSQL Documentation
- ☐ <https://www.postgresql.org/docs>
- ☐ LangChain Documentation
- ☐ <https://docs.langchain.com>